

AJAYA DAHAL

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SUMMARY

Sr. Silicon Design Engineer at AMD specializing in post-silicon root-cause analysis of x86 data center CPUs (SDC/MCE), stress-to-fail characterization across PVT corners, and AI-driven failure analysis tooling (RAG/NL2SQL). Background in high-speed Ethernet (10G–800G) and FPGA-based test platforms, with hands-on lab skills in signal integrity (GTY/GTM transceivers up to 58 Gbps), power integrity (di/dt-induced voltage noise), and embedded systems (PetaLinux/Yocto, bare-metal, kernel drivers).

PROFESSIONAL EXPERIENCE

- Sr. Silicon Design Engineer, AMD (x86 Data Center CPU Post-Silicon Debug)** Austin, TX | Jan 2026 – Present
- Lead architectural root-cause analysis of Silent Data Corruption (SDC) and Machine Check Exceptions (MCE) on next-gen x86 data center CPUs, identifying failure mechanisms at the microarchitecture level.
 - Characterized FP Physical Register File (FP-PRF) stability by architecting stress-to-fail campaigns using high-entropy binaries (FP_Deluge) to induce maximum di/dt voltage noise and expose power integrity margins.
 - Architected “Silicon Trace,” an AI-driven failure analysis platform using RAG/NL2SQL to automate telemetry correlation, reducing manual debug cycles for complex MCE signatures.
 - Developed and released 6 debug-analysis agents that let the team and partners analyze and document debug data from JIRA tickets and test-program spreadsheets; received the AMD SIP Infinity Spotlight: Innovation award (2026).
 - Reproduced customer-reported failures across PVT (process, voltage, temperature) corners using the AFHC framework; serve as primary bridge between Customer Engineering and Silicon Design (IP) teams.
- Sr. Embedded Product Application Engineer, AMD (Adaptive & Embedded Computing Group, Xilinx)** Remote | Jan 2023 – Dec 2025
- Technical lead for 15+ Ethernet reference designs from 10 Mbps TEMAC to 800G DCMAC, integrating DCMAC, MRMAC and CMAC hard IP with DMA subsystems on Versal ACAP and Zynq UltraScale+ MPSoC; built Python automation for test execution, data extraction and reporting.
 - Troubleshoot GTM/GTY/GTH/GTYP transceiver issues and optimized eye diagrams via IBERT at 28–58 Gbps/lane; full-stack debug from RTL (Verilog) through PetaLinux/Yocto and bare-metal BSPs (AXI NoC data paths, PS-PL interfaces, Ethernet DMA).
 - Authored AMD Answer Records and tutorials (MRMAC 4x25G tutorial series, DCMAC Survival Guide) resolving customer blockers on Versal (VMK180, VPK120, VCK190) and Zynq UltraScale+ (ZCU102, ZCU106, ZCU111).
 - Recognized with 8 AMD Spotlight Awards (2023–2025), including WTS NA Field Recognition 2025 and PAE Team Player 2024, for customer-critical MRMAC, CMAC, QSGMII, PS GEM TSU and AXI 2.5G work.
- Graduate Research Assistant, Mississippi State University (ECE, with CAVS)** Starkville, MS | Aug 2019 – Jan 2024
- Built SDR-based RF sensing systems (contraband detection, spectrum scanning, Wi-Fi HAR with ML) and autonomous vehicle perception (LIDAR-camera fusion, TensorFlow). 12 publications, 45+ citations. FAA Part 107 certified.
- PCB Design Engineer Co-op, Hunter Engineering Company** Raymond, MS | Jan 2020 – Jan 2022
- Designed end-to-end mixed-signal PCB functional testers with custom test boards, LabVIEW GUIs, C/C++ backends, and camera-based defect detection (OpenCV), integrated with Aegis Factory Logix MES.

SELECTED PROJECTS

- WiFi HAR on the Versal AI Engine — M.S. Thesis on Silicon** 2026
- Streams WiFi CSI from a Raspberry Pi (nexmon) into a Versal VCK190; the PL parses UDP/CSI packets and the AI Engine runs STFT activity, breathing and fall features, bit-accurate on silicon.
- 72ns Gateway — Versal ACAP HFT Network Accelerator** Dec 2025
- Sub-100ns packet processing on Versal: wire-speed UDP filtering, Gray-code CDC, NASDAQ ITCH 5.0 parsing (9 cycles @ 125 MHz, 1 Gbps).

EDUCATION

- M.S. in Electrical & Computer Engineering, Mississippi State University** | GPA 4.0/4.0 | Thesis: SDR-Based Sensing May 2024
- B.S. in Electrical & Computer Engineering, Mississippi State University** | GPA 4.0/4.0 | Focus: FPGA, Digital & Embedded Systems Dec 2022

TECHNICAL SKILLS

Silicon Validation: Post-silicon debug (SDC/MCE), stress-to-fail characterization, PVT corner testing, AFHC, di/dt voltage noise analysis
FPGA & RTL: Vivado, Vitis, Verilog, Vitis HLS, ChipScope/IBERT, ModelSim, JTAG, Versal ACAP, Zynq UltraScale+, RFSoc
Protocols: 800G/400G DCMAC, 100G MRMAC/CMAC, 10G/25G XXV Ethernet, GTY/GTM/GTYP SerDes (28–58 Gbps), DDR, PCIe, AXI, SPI
Embedded & OS: PetaLinux, Yocto, bare-metal BSP, Linux kernel/driver development, C, C++, Device Tree
Automation & AI: Python (FastAPI/Streamlit), MATLAB, RAG/NL2SQL, TensorFlow, Docker, Jenkins CI/CD
Lab & Measurement: Oscilloscopes, Logic Analyzers, BERT, Eye diagrams, Signal Integrity, USRP (x310/N310), VNA

CERTIFICATIONS & AWARDS

Certification: FAA Part 107 Remote Pilot | **Training:** NVIDIA RDMA Programming, Verilog HDL, Linux Device Drivers, Yocto, LPIC-2 prep
Awards: 9 AMD Spotlight Awards (2023–2026), incl. SIP Infinity Innovation 2026, WTS NA Field Recognition 2025, PAE Team Player 2024 | NXP HoverGames Bonus Prize 2021 | 12 publications (45+ citations)